



North South University

Department of Electrical and Computer Engineering

CSE 215L (Programming Language II Lab)

Lab 1: Introduction, Environment Setup, Console I/O

Objective

- **Introduction to Programming Environment:** Familiarize students with the course's programming environment and IDE.
- **Setting Up Development Tools:** Guide students in installing and configuring essential programming tools such as the JDK, compiler, and debugger.
- **Understanding Console Input and Output:** Introduce basic console-based input and output operations in Java.
- **Variable Declaration and Initialization:** Introduce variables and their declaration and initialization in Java.
- **Data Types and their Usage:** Discuss fundamental data types with simple practical examples.
- **User Interaction through Console I/O:** Enable students to create programs that interact with users through console input.

Introduction to Java

Java is a general-purpose, object-oriented programming language known for its portability and extensive standard library. Java source code is compiled into bytecode, which is executed by the Java Virtual Machine (JVM). This supports the principle of *Write Once, Run Anywhere*.

Java is widely used in application development, backend systems, enterprise software, and many other areas of computing.

Source Code → Compiler → Bytecode → JVM → Output

Instructions for Setting up the Environment of Java

If you are planning to develop and run Java programs, follow these steps:

- Install the **Java Development Kit (JDK)**. The JDK contains the compiler and other tools required for Java development.
- The **Java Virtual Machine (JVM)** executes Java bytecode. For modern Java development, installing the JDK is sufficient; a separate JRE installation is generally not required.
- Java can be written using a text editor and compiled from the command line. An **IDE** provides a convenient environment for writing, compiling, running, and debugging programs.
- For this course, **Apache NetBeans** may be used as the IDE.
- Download an appropriate JDK from:

<https://www.oracle.com/java/technologies/downloads/>

- Install the JDK and verify the installation:

```
java --version
javac --version
```

- Download Apache NetBeans from:

<https://netbeans.apache.org/>

Console I/O

Programming requires both input and output. In this laboratory, **Scanner** is used for console input and **System.out** for console output.

First import Scanner:

```
import java.util.Scanner;
```

Create a Scanner object:

```
Scanner input = new Scanner(System.in);
```

Here, **System.in** represents standard input and **input** is the Scanner object.

The commonly used Scanner methods are:

Method	Description
<code>nextByte()</code>	Reads a byte.
<code>nextShort()</code>	Reads a short.
<code>nextInt()</code>	Reads an int.
<code>nextLong()</code>	Reads a long.
<code>nextFloat()</code>	Reads a float.
<code>nextDouble()</code>	Reads a double.
<code>next()</code>	Reads the next token until whitespace.
<code>nextLine()</code>	Reads an entire line of text.
<code>nextBoolean()</code>	Reads true or false.

Among these, `nextInt()`, `nextDouble()`, `nextFloat()`, `next()`, and `nextLine()` will be used frequently.

Reading Input

To read a String:

```
String statement = input.nextLine();
```

To read an integer:

```
int points = input.nextInt();
```

To read a decimal number:

```
double balance = input.nextDouble();
```

Printing Output

Output can be produced using:

```
System.out.println(<something>);
```

`println()` can print strings, variables, numbers, or combinations of them. The cursor moves to the next line after printing.

Example:

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

The `main()` method is the starting point of a standard Java application.

Variables and Data Types

A variable is a named storage location used to hold a value.

```
int points = 95;
double balance = 500.50;
```

```
String name = "Faiyaz";
```

Common Java data types:

Type	Category	Example
int	Integer	int points = 95;
double	Decimal	double balance = 500.50;
float	Decimal	float temp = 36.5f;
char	Character	char grade = 'A';
boolean	True/False	boolean passed = true;
String	Text	String name = "Faiyaz";

String is a reference type rather than a primitive type, but is commonly used for text input.

Classwork

This is the first laboratory; therefore, the classwork focuses on basic program execution.

1. Write and execute the following Hello World program:

```
public class HelloWorld {  
  
    public static void main(String[] args) {  
        System.out.println("Hello World!");  
    }  
}
```

2. Create a String variable named `statement`, assign “Hello World!” to it, and print the variable.

```
String statement = "Hello World!";  
System.out.println(statement);
```

Homework

Do the following tasks:

1. Create a String variable `name` and assign the value of the user’s input.
2. Create an `int` variable `points` and assign the value of the user’s input.
3. Create a `double` variable `balance` and assign the value of the user’s input.
4. Print these variables in **ONE LINE**.